

Solvency Field Theory V12.3: Gravity Trajectory Prevalidation Closure

Subtitle: From One-Speed Settlement Geometry to a Frozen Blind Prediction Protocol

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Version: SFT V12.3 Gravity Trajectory

Abstract

This paper records the SFT Gravity Trajectory audit sequence GRAV_TRAJ_01 through GRAV_TRAJ_13. The purpose of the sequence is not to fit galaxy data, solve clusters, predict kSZ signals, or complete nonlinear gravity. Its purpose is narrower and stricter: to construct and freeze a pre-data mechanism chain connecting the one-speed settlement picture of gravity to a blind prediction protocol for future validation.

The sequence begins with a helical one-speed source kernel, verifies compatibility with the V11.13 one-speed weak-field metric closure, extends the source rule continuously across matter and radiation through a $\rho + 3p$ operation-density structure, selects an acceleration-triggered far-field settlement handoff, derives the settlement acceleration scale $a_S = cH/(2\pi)$, constrains the 2π phase-cycle coupling, identifies H as a background settlement update rate, and derives $H(t) = c/R(t)$ from no-overwrite settlement-radius dynamics. It then introduces an environmental deformation factor $E = \sqrt{DB}$, where D is a settlement-density channel and B is a boundary-connectivity channel, proves that E may enter only the far-field handoff and not the local metric/source sector, selects physical proxy classes for D and B , and finally freezes a no-retuning blind prediction protocol.

The final result is a sealed prevalidation recipe with protocol hash:

f38b5295daa54a25c6ed1730b204d9c59a20ef88ef4661928aee6787463749cc

The locked protocol computes ρ_{set} from the operation-density proxy $\rho_b + 3p_b/c^2$, with cold-system fallback $\rho_{\text{set}} = \rho_b$; computes D by background-normalized smoothing; computes B from predeclared source-graph boundary conductance; uses $r_0 = \sqrt{GM/a_S}$ with fixed scale fractions for smoothing and regions; and predicts $E = \sqrt{DB}$, $a_{\text{eff}} = Ea_S$, $r_t = r_0/\sqrt{E}$, and the deep-regime $\sqrt{E}$ deviation. Observed a_0 , rotation-curve residuals, SPARC/RAR residuals, cluster residuals, kSZ targets, fitted dark-halo parameters, post-hoc environment labels, best-fit kernel lengths, and per-object exponents are explicitly forbidden inputs. Gate 14 and later validation tests are reserved for appendices and must use the frozen recipe or declare a fork.

1. Scope and Claim Boundary

This paper is a mechanism-construction and protocol-freeze paper. It is not an observational validation paper.

The claims earned here are:

1. SFT has a staged gravity trajectory connecting one-speed settlement source geometry to a pre-data far-field prediction protocol.
2. The local weak-field metric closure from the V11.13 gravity branch is preserved.
3. The environmental factor E is confined to the far-field settlement handoff.
4. A no-retuning blind prediction protocol has been frozen before data-facing validation.

The claims not made here are equally important:

1. This paper does not claim a SPARC/RAR fit.
2. This paper does not claim a cluster solution.
3. This paper does not claim a kSZ prediction.
4. This paper does not derive final nonlinear gravitational field equations.
5. This paper does not claim that the physical D and B proxies are final truth.
6. This paper does not claim that SFT has solved quantum gravity.

Gate 14 and later validation work should be treated as appendices or companion validation notes, not as part of the pre-data construction sequence.

2. Why This Is V12.3, Not V11.14

The result belongs naturally to **SFT V12.3** rather than V11.14.

V11.13 established the weak-field settlement-gravity skeleton and localized the absolute gravitational scale problem. V12.1 supplied the One-Shatter initial-condition closure. V12.2 supplied the quantum-gravity root-scale selector candidate. The present sequence uses those results as upstream context, but it develops a new downstream gravity trajectory: source geometry, far-field settlement handoff, environment factors, and a blind validation protocol.

Therefore the clean naming is:

SFT V12.3 Gravity Trajectory Prevalidation Closure
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This avoids implying that the paper is merely a patch to V11.13, while preserving the continuity of the V11.13 gravity branch.

3. Core Language

The central SFT gravity reading used throughout this sequence is:

Gravity is the coarse-grained settlement geometry of maintained one-speed structures.

Equivalently:

Time dilation is the local-value face of settlement lag; attraction is the gradient face; spatial curvature is the reciprocal one-speed closure face.

The phrase “one-speed” does not posit an aether, lattice, foam, or background medium. It means that real particles and fields preserve and spend a one-speed causal budget. Rest mass appears as folded internal budget maintenance rather than literal corpuscles moving through space at c .

4. Summary of the Gate Chain

The gravity trajectory now has thirteen prevalidation gates:

- 01 : helix source kernel
- 02 : one-speed metric closure
- 03 : $\rho + 3p$ source continuity
- 04 : acceleration-triggered settlement handoff
- 05 : $a_S = cH/(2\pi)$
- 06 : 2π phase-cycle selector
- 07 : H as background settlement update rate
- 08 : $H(t) = c/R(t)$
- 09 : $E = \sqrt{DB}$
- 10 : computable synthetic D/B estimators
- 11 : E far-field only; local $FB = 1$ preserved
- 12 : physical D/B proxy bundle
- 13 : blind prediction protocol freeze

After Gate 13, the next stage is data-facing validation:

- 14+ : SPARC/RAR/cluster/kSZ validation appendices.
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5. GRAV_TRAJ_01: One-Helix Lag Field Audit

Gate 01 modeled a rest-frame one-speed internal circulation as a time-averaged ring source with Green kernel $1/|x - x'|$. The goal was not to derive full gravity, but to test whether a one-speed helical source can seed the correct coarse-grained lag geometry.

The verdict was:

ONE_HELIX_RING_SOURCE_PASSES_DIRECTIONAL_LAG_SEED_AUDIT

The spherical average recovers the Newtonian exterior $1/r$ behavior outside the ring. The leading angular correction has the expected equatorial-positive and polar-negative quadrupolar form:

$$\frac{\Lambda}{\Lambda_0} = 1 + \epsilon(r) \left(\sin^2 \theta - \frac{2}{3} \right),$$

with asymptotic behavior:

$$\epsilon(r) \rightarrow 0.75 \left(\frac{a}{r} \right)^2.$$

Numerically, fit-to-prediction ratios were approximately unity over the tested range. Gate 01 therefore supplies a source-kernel seed for directional lag, not a full gravitational theory.

6. GRAV_TRAJ_02: Source-to-Metric Closure Audit

Gate 02 tested whether the Gate 01 source kernel is compatible with the V11.13 weak-field metric closure.

The local metric scaffold is:

$$F = e^{-\phi}, \quad B = e^{\alpha\phi}.$$

The time and radial metric components expand as:

$$g_{tt} = -F^2 = -1 + 2\phi - 2\phi^2 + \dots,$$

$$g_{rr} = B^2 = 1 + 2\alpha\phi + \dots.$$

The corresponding PPN markers are:

$$\gamma = \alpha,$$

$$\beta = 1.$$

The one-speed reciprocal closure selects:

$$\alpha = 1,$$

so that:

$$FB = 1,$$

$$\gamma = 1,$$

$$\beta = 1.$$

The verdict was:

HELIX_SOURCE_COMPATIBLE_WITH_V11_13_ONE_SPEED_METRIC_CLOSURE

Gate 02 is the local weak-field guardrail for the rest of the sequence.

7. GRAV_TRAJ_03: Matter-Radiation Source Continuity Audit

Gate 03 tested whether the source rule can pass continuously from cold matter to radiation.

The selected source structure is the active operation-density form:

$$\rho_{\text{ops}} \propto \rho + 3p.$$

With equation-of-state parameter:

$$w = \frac{p}{\rho},$$

this gives source factor:

$$1 + 3w.$$

In one-speed budget language, the isotropic spatial budget is:

$$b = \langle \beta_{\text{space}}^2 \rangle,$$

and the source factor becomes:

$$1 + b.$$

Cold matter has:

$$b = 0 \Rightarrow 1,$$

while radiation has:

$$b = 1 \Rightarrow 2.$$

Anisotropic components preserve the trace source while producing traceless directional structure. The verdict was:

MATTER_RADIATION_ONE_SPEED_SOURCE_CONTINUITY_PASSES

This gate is later reused in Gate 12 when selecting the physical D -channel proxy.

8. GRAV_TRAJ_04: Settlement Transition Audit

Gate 04 tested the structure of the far-field settlement handoff. The central question was whether SFT should reproduce MOND exactly, or whether it should produce a MOND-like transition with allowed deviations.

The selected reading is:

SFT is MOND-like, not MOND.

A fixed universal length scale can mimic a flat curve for one mass, but it fails the baryonic Tully-Fisher scaling because it gives:

$$v^4 \propto M^2.$$

The acceleration-triggered transition radius:

$$r_t = \sqrt{\frac{GM}{a_S}}$$

instead gives the far-regime relation:

$$g \approx \sqrt{g_N a_S},$$

and therefore:

$$v_{\text{flat}}^4 \propto GM a_S.$$

Gate 04 also introduced the possibility of an environment factor:

$$a_{\text{eff}} = E a_S,$$

so that:

$$r_t \propto \frac{1}{\sqrt{E}},$$

and the deep-regime amplitude shifts with E .

The verdict was:

This gate did not use observed a_0 , galaxy data, SPARC/RAR data, or kSZ data.

9. GRAV_TRAJ_05: Settlement Acceleration Scale Audit

Gate 05 tested the no-fit acceleration scale:

$$a_S = \frac{cH_0}{2\pi} = \frac{c^2}{2\pi R_H}.$$

Using fiducial $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, the audit gives:

$$a_S = 1.0824013602391998 \times 10^{-10} \text{ m/s}^2.$$

The invariant checked across tested radii was:

$$\frac{a_S R}{c^2} = \frac{1}{2\pi}.$$

The verdict was:

This is a constraint audit only. It is not a first-principles derivation of observed a_0 , not a SPARC fit, and not a kSZ prediction.

10. GRAV_TRAJ_06: Phase-Cycle Coupling Audit

Gate 06 tested the origin of the 2π factor in:

$$a_S = \frac{cH}{2\pi}.$$

The selected coupling was:

$$K = 2\pi.$$

The operational clauses selecting 2π were:

1. phase identity,
2. cycle-versus-radian accounting,
3. root-area cycle relation,
4. one-speed horizon settlement.

Controls such as $K = 1$, $K = \pi$, $K = 4\pi$, and target-inverted exact values were rejected or quarantined.

The verdict was:

PHASE_CYCLE_2PI_COUPLING_SURVIVES_OPERATIONAL_SELECTOR_AUDIT

Remaining debts after this gate included deeper microscopic derivation of the phase-cycle coupling and the environment factor.

11. GRAV_TRAJ_07: H as Settlement Update Rate

Gate 07 tested whether H can be interpreted as the background settlement update rate.

The selected identity was:

$$H = \frac{\dot{a}}{a} = \frac{c}{R_H}.$$

With fiducial $H_0 = 70$, the audit gives:

$$\begin{aligned} H &= 2.2685455026110557 \times 10^{-18} \text{ s}^{-1}, \\ R_H &= 13.96846030972556 \text{ Gly}, \\ a_S &= \frac{cH}{2\pi} = 1.0824013602391998 \times 10^{-10} \text{ m/s}^2. \end{aligned}$$

The verdict was:

H_SURVIVES_AS_BACKGROUND_SETTLEMENT_UPDATE_RATE_CANDIDATE

This gate did not derive H microscopically or validate any data comparison. It identified the role of H in the settlement-rate chain.

12. GRAV_TRAJ_08: $H(t) = c/R(t)$ from No-Overwrite Settlement Radius

Gate 08 tested whether the time-dependent update rate follows from no-overwrite settlement-radius dynamics.

The governing assumption is:

$$\frac{dR}{dt} = c.$$

Then the fractional ledger update rate is:

$$H_{\text{set}}(t) = \frac{d \ln R}{dt} = \frac{c}{R(t)}.$$

The corresponding acceleration scale is:

$$a_S(t) = \frac{cH_{\text{set}}(t)}{2\pi} = \frac{c^2}{2\pi R(t)}.$$

The integral closure is:

$$\int H_{\text{set}} dt = \ln \left(\frac{R_f}{R_i} \right).$$

The patched numerical verdict was:

H_OF_T_SURVIVES_AS_NO_OVERWRITE_SETTLEMENT_RADIUS_RATE_NUMERIC_TOLERANCE_

For the Gate 11R recombination-to-present path:

$$\begin{aligned} R_i &= 0.624698 \text{ Mly}, \\ R_f &= 13968.46031 \text{ Mly}, \\ \frac{R_f}{R_i} &= 22360.333453, \\ \ln(R_f/R_i) &= 10.015044. \end{aligned}$$

For the V12.1 locked recombination-to-present path:

$$\begin{aligned} R_i &= 0.628104 \text{ Mly}, \\ \frac{R_f}{R_i} &= 22239.075536, \\ \ln(R_f/R_i) &= 10.009606. \end{aligned}$$

Gate 08 constrains $H(t)$ as a no-overwrite settlement-radius rate, while leaving open the microscopic derivation of $R(t)$, the mapping to FRW observables, and the environment factor.

13. GRAV_TRAJ_09: Environment Factor $E = \sqrt{DB}$

Gate 09 introduced the environment factor through two dimensionless channels:

$$\begin{aligned} D &= \frac{\rho_{\text{set}}}{\bar{\rho}_{\text{set}}}, \\ B &= \frac{C_{\text{boundary}}}{\bar{C}_{\text{boundary}}}. \end{aligned}$$

The selected no-bias equal-log law was:

$$\ln E = \frac{\ln D + \ln B}{2},$$

or:

$$E = \sqrt{DB}.$$

The verdict was:

ENVIRONMENT_FACTOR_E_SURVIVES_NO_BIAS_SETTLEMENT_DENSITY_CONNECTIVITY_AUDI

The equal-log selector scored 1.000. Controls were rejected as follows:

- arithmetic mean failed reciprocal log-capacity neutrality,
- density-only failed exchange symmetry,
- connectivity-only failed exchange symmetry,
- free-alpha log power was rejected as tunable,
- target residual fitting was forbidden.

With:

$$a_{\text{eff}} = E a_S,$$

one obtains:

$$r_t(E) = \frac{r_t(1)}{\sqrt{E}},$$

and the deep-regime amplitude shift:

$$\frac{g_{\text{env}}}{g_{\text{universal}}} = \sqrt{E}.$$

Synthetic examples showed:

$$E = 0.5 \Rightarrow \sqrt{E} = 0.7071,$$

$$E = 2 \Rightarrow \sqrt{E} = 1.4142,$$

$$E = 2.8284 \Rightarrow \sqrt{E} = 1.6818.$$

Compensated environments such as $D = 2, B = 0.5$ give $E = 1$, preserving background-like behavior. Gate 09 therefore provides a non-universal MOND-like deviation channel without fitting residuals.

14. GRAV_TRAJ_10: Computable Synthetic D/B Estimators

Gate 10 converted the symbolic D and B channels into computable synthetic estimators before observational comparison.

The selected synthetic settlement-density estimator was:

$$D(x) = \frac{\rho_{\text{set}}(x)}{\langle \rho_{\text{set}} \rangle_{\text{background}}},$$

where ρ_{set} is a Gaussian-smoothed settlement field.

The selected boundary-connectivity estimator was:

$$B(S) = \frac{C_{\text{boundary}}(S)}{\langle C_{\text{boundary}} \rangle_{\text{background}}},$$

where C_{boundary} is graph cut conductance per node.

The verdict was:

COMPUTABLE_D_B_ESTIMATORS_SURVIVE_FROZEN_SYNTHETIC_GRAPH_FIELD_AUDIT

Parameters in the audit were:

$$\ell_D = 0.1,$$

$$\ell_B = 0.16,$$

$$R_{\text{region}} = 0.18.$$

Dense-connected synthetic systems raised E , underconnected voids lowered E , and compensated cases demonstrated that boundary connectivity can reduce or enhance a density-only expectation. Gate 10 does not identify final physical observables; it proves that the environment channel can be computed from frozen graph/field rules before data fitting.

15. GRAV_TRAJ_11: Far-Field E Attachment and Local Metric Preservation

Gate 11 tested where the environment factor may enter the gravity trajectory.

The surviving attachment is:

$$a_{\text{eff}} = E a_S(t),$$

$$r_t(E) = \sqrt{\frac{GM}{Ea_S}}.$$

The local metric remains unchanged:

$$\begin{aligned} F &= e^{-\phi}, \\ B_{\text{metric}} &= e^{+\phi}, \\ FB &= 1, \end{aligned}$$

with:

$$\begin{aligned} \gamma &= 1, \\ \beta &= 1. \end{aligned}$$

The audit kernel was:

$$\frac{g}{g_N} = \sqrt{1 + \left(\frac{r}{r_t}\right)^2}.$$

Near field:

$$r \ll r_t \Rightarrow \frac{g}{g_N} \rightarrow 1.$$

Deep field:

$$r \gg r_t \Rightarrow g \rightarrow \sqrt{g_N E a_S}.$$

The verdict was:

FAR_FIELD_ONLY_E_ATTACHMENT_PRESERVES_LOCAL_METRIC_CLOSURE

Controls that placed E inside the metric potential, spatial closure alpha, local source mass, additive acceleration floor, or target-fitted residuals failed or were forbidden. The rule is now:

E may move the handoff radius, but may not touch the local metric/source sector.

16. GRAV_TRAJ_12: Physical D/B Proxy Bundle

Gate 12 selected physical proxy classes for the D and B channels while enforcing the Gate 11 guardrail.

The selected settlement-density proxy is:

$$\rho_{\text{set}} \propto \rho_b + \frac{3p_b}{c^2}.$$

For cold systems where pressure is unavailable or negligible:

$$\rho_{\text{set}} \rightarrow \rho_b.$$

The selected boundary-connectivity proxy is:

$$C_{\text{boundary}} = \text{source-graph cut conductance per node.}$$

The selected scale is:

$$r_0 = \sqrt{\frac{GM}{a_S}}.$$

The verdict was:

PHYSICAL_D_B_PROXY_BUNDLE_SELECTED_WITH_GATE11_GUARDRAIL

Rejected or quarantined controls included:

- stellar-light-only proxy,
- fitted dark-halo density proxy,
- rotation-curve residual proxy,
- nearest-neighbor density as a true B proxy,
- visual morphology labels,
- residual-fit connectivity,
- fixed 10 kpc scale,
- best-fit kernel scale.

Scale examples at $a_S = 1.0824013602391998 \times 10^{-10} \text{ m/s}^2$ were:

$$M = 10^8 M_{\odot} \Rightarrow r_0 = 0.358854 \text{ kpc},$$

$$M = 10^{10} M_{\odot} \Rightarrow r_0 = 3.588539 \text{ kpc},$$

$$M = 10^{11} M_{\odot} \Rightarrow r_0 = 11.347957 \text{ kpc},$$

$$M = 10^{14} M_{\odot} \Rightarrow r_0 = 358.853903 \text{ kpc}.$$

Gate 12 produces a complete proxy bundle but not an observational validation.

17. GRAV_TRAJ_13: Blind Prediction Protocol Freeze

Gate 13 froze the blind prediction protocol and wrote a no-retuning lockbox.

The protocol version is:

GRAV_TRAJ_13_BLIND_PROTOCOL_FREEZE_v1

The protocol hash is:

```
f38b5295daa54a25c6ed1730b204d9c59a20ef88ef4661928aee6787463749cc
```

The locked protocol computes:

$$\rho_{\text{set}} = \rho_b + \frac{3p_b}{c^2},$$

with cold fallback:

$$\rho_{\text{set}} = \rho_b.$$

Then:

$$D = \frac{\rho_{\text{set}}}{\langle \rho_{\text{set}} \rangle_{\text{background}}},$$

$$B = \frac{C_{\text{boundary}}}{\langle C_{\text{boundary}} \rangle_{\text{background}}},$$

$$E = \sqrt{DB},$$

$$a_{\text{eff}} = E a_S,$$

$$r_t = \frac{r_0}{\sqrt{E}},$$

$$r_0 = \sqrt{\frac{GM}{a_S}}.$$

The scale fractions are locked:

$$\ell_D = 0.1r_0,$$

$$\ell_B = 0.16r_0,$$

$$R_{\text{region}} = 0.18r_0.$$

Forbidden inputs include:

- observed a_0 target,
- rotation-curve residuals,
- SPARC/RAR fit residuals,
- cluster mass discrepancy residuals,
- kSZ residuals,
- dark-halo best-fit parameters,
- post-hoc environment labels,
- best-fit kernel lengths,
- per-object free exponents or offsets.

The no-retuning firewall locks:

1. protocol hash before validation,
2. no kernel refit,
3. no E -exponent refit,
4. no local metric modification,
5. no target inputs,
6. sealed prediction table before validation.

The verdict was:

BLIND_PREDICTION_PROTOCOL_FROZEN_READY_FOR_DATA_FACING_GATE14

Gate 13 completes the pre-data mechanism chain.

18. Synthetic Smoke Predictions from the Frozen Protocol

The Gate 13 smoke table is not validation data. It only proves that the equations generate finite prediction rows without target inputs.

For a background dwarf:

$$\begin{aligned}M &= 10^8 M_{\odot}, \\D &= 1, \\B &= 1, \\E &= 1, \\r_0 = r_t &= 0.358854 \text{ kpc}.\end{aligned}$$

For a dense-connected galaxy:

$$\begin{aligned}M &= 10^{10} M_{\odot}, \\D &= 2, \\B &= 2, \\E &= 2, \\\sqrt{E} &= 1.414214, \\r_0 &= 3.588539 \text{ kpc}, \\r_t &= 2.537480 \text{ kpc}.\end{aligned}$$

For a large screened system:

$$M = 10^{11} M_{\odot},$$

$$D = 4,$$

$$B = 0.25,$$

$$E = 1,$$

$$r_0 = r_t = 11.347957 \text{ kpc}.$$

For a void-like system:

$$M = 10^{10} M_{\odot},$$

$$D = 0.5,$$

$$B = 0.5,$$

$$E = 0.5,$$

$$\sqrt{E} = 0.707107,$$

$$r_t = 5.074961 \text{ kpc}.$$

For a connected cluster-like system:

$$M = 10^{14} M_{\odot},$$

$$D = 4,$$

$$B = 2,$$

$$E = 2.828427,$$

$$\sqrt{E} = 1.681793,$$

$$r_0 = 358.853903 \text{ kpc},$$

$$r_t = 213.375807 \text{ kpc}.$$

19. Resulting Prediction Logic

The frozen protocol predicts three environment-facing quantities before validation:

$$E = \sqrt{DB},$$

$$\sqrt{E},$$

$$r_t = \frac{r_0}{\sqrt{E}}.$$

A high-density, high-connectivity environment has $E > 1$, which decreases the transition radius and raises the deep-regime amplitude.

A low-density, low-connectivity environment has $E < 1$, which increases the transition radius and lowers the deep-regime amplitude.

A compensated environment with $DB = 1$ has $E = 1$, even if density or connectivity alone is non-background. This is the central reason B cannot be replaced by a second density proxy.

The local metric remains unchanged in all cases.

20. Discussion

The GRAV_TRAJ sequence is significant because it separates mechanism construction from observational validation.

Before this sequence, the SFT gravity branch had three open risks:

1. the source geometry could have been disconnected from metric closure,
2. the MOND-like far-field behavior could have been a hidden fit target,
3. environment dependence could have corrupted local GR-like behavior.

The thirteen gates address these risks in sequence.

Gates 01 and 02 connect source geometry to weak-field one-speed metric closure. Gate 03 preserves matter-radiation source continuity. Gates 04 through 08 build the settlement acceleration and update-rate chain. Gates 09 through 12 construct the environment channel and constrain it to a physical no-target proxy bundle. Gate 13 freezes the protocol and forbids retuning.

The result is not empirical success. It is readiness for blind validation.

That distinction is essential. A data-facing comparison that changes the graph, kernel, exponent, background sample, or environment definition after seeing residuals is not Gate 14 validation of this protocol. It is a fork.

21. Gate 14 as Appendix

Gate 14 should be treated as an appendix or companion validation note.

The first validation appendix should not immediately claim a SPARC fit. It should begin with:

Gate 14A: validation readiness and protocol hash check

Gate 14A should verify:

1. the protocol hash is unchanged,
2. the input dataset is explicitly declared,
3. the allowed fields are present,
4. forbidden target fields are excluded from the prediction computation,
5. the graph construction can be run as frozen,
6. the prediction table is sealed before residual comparison.

Only after Gate 14A should later appendices perform:

Gate 14B: real graph implementation

Gate 14C: first SPARC/RAR-facing comparison

and then cluster or kSZ validation appendices.

22. Conclusion

SFT V12.3 completes a pre-data gravity trajectory from one-speed settlement source geometry to a frozen blind prediction protocol.

The final locked equations are:

$$\rho_{\text{set}} = \rho_b + \frac{3p_b}{c^2},$$

with cold fallback:

$$\begin{aligned} \rho_{\text{set}} &= \rho_b, \\ D &= \frac{\rho_{\text{set}}}{\langle \rho_{\text{set}} \rangle_{\text{background}}}, \\ B &= \frac{C_{\text{boundary}}}{\langle C_{\text{boundary}} \rangle_{\text{background}}}, \\ E &= \sqrt{DB}, \\ a_{\text{eff}} &= E a_S, \\ r_0 &= \sqrt{\frac{GM}{a_S}}, \\ r_t &= \frac{r_0}{\sqrt{E}}. \end{aligned}$$

The local metric guardrail remains:

$$F = e^{-\phi},$$

$$B_{\text{metric}} = e^{+\phi},$$

$$FB = 1,$$

$$\gamma = 1,$$

$$\beta = 1.$$

The protocol hash is:

f38b5295daa54a25c6ed1730b204d9c59a20ef88ef4661928aee6787463749cc

The status is therefore:

pre-data gravity trajectory closed; blind validation ready.

Not:

observational validation complete.

Gate 14 and later validation work must be appended using the sealed protocol or explicitly declared as a fork.

Appendix A. Gate Verdicts

Gate	Verdict
GRAV_TRAJ_01	ONE_HELIX_RING_SOURCE_PASSES_DIRECTIONAL_LAG_SEED_AUDIT
GRAV_TRAJ_02	HELIX_SOURCE_COMPATIBLE_WITH_V11_13_ONE_SPEED_METRIC_CLOSURE
GRAV_TRAJ_03	MATTER_RADIATION_ONE_SPEED_SOURCE_CONTINUITY_PASSES
GRAV_TRAJ_04	ACCELERATION_TRIGGERED_SETTLEMENT_HANDOFF_SELECTED_AS_MOND_LIKE_NOT_MOND_STRUCTURE

Gate	Verdict
GRAV_TRAJ_05	HORIZON_PHASE_CYCLE_SETTLEMENT_ACCELERATION_SURVIVES_NO_FIT_CONSTRAINT_AUDIT
GRAV_TRAJ_06	PHASE_CYCLE_2PI_COUPLING_SURVIVES_OPERATIONAL_SELECTOR_AUDIT
GRAV_TRAJ_07	H_SURVIVES_AS_BACKGROUND_SETTLEMENT_UPDATE_RATE_CANDIDATE
GRAV_TRAJ_08	H_OF_T_SURVIVES_AS_NO_OVERWRITE_SETTLEMENT_RADIUS_RATE_NUMERIC_TOLERANCE_PATCHED
GRAV_TRAJ_09	ENVIRONMENT_FACTOR_E_SURVIVES_NO_BIAS_SETTLEMENT_DENSITY_CONNECTIVITY_AUDIT
GRAV_TRAJ_10	COMPUTABLE_D_B_ESTIMATORS_SURVIVE_FROZEN_SYNTHETIC_GRAPH_FIELD_AUDIT
GRAV_TRAJ_11	FAR_FIELD_ONLY_E_ATTACHMENT_PRESERVES_LOCAL_METRIC_CLOSURE
GRAV_TRAJ_12	PHYSICAL_D_B_PROXY_BUNDLE_SELECTED_WITH_GATE11_GUARDRAIL
GRAV_TRAJ_13	BLIND_PREDICTION_PROTOCOL_FROZEN_READY_FOR_DATA_FACING_GATE14

Appendix B. Protocol Hash

f38b5295daa54a25c6ed1730b204d9c59a20ef88ef4661928aee6787463749cc

Any future validation appendix must either use this protocol hash or explicitly declare a new fork.

Appendix C. Gate 14 Reserved Validation Appendix

Gate 14 is reserved for validation readiness and data-facing tests. It should not be merged into the core mechanism paper unless the validation is complete and the protocol hash remains unchanged.

Recommended order:

1. Gate 14A: readiness and hash check,
2. Gate 14B: real graph implementation,
3. Gate 14C: SPARC/RAR-facing comparison,
4. Gate 14D: cluster-facing comparison,
5. Gate 14E: kSZ/large-scale flow-facing comparison.

All validation appendices must report failures honestly and must not retune the frozen protocol after seeing validation targets.